

SpaceX – why it is crap and AI in space is too expensive.

First of all, SpaceX's valuation is based, of course, on three models: the space industry, the xAI platform, and Starlink. I will try to convince you why xAI in space is totally bullshit.

The major thing which is, of course, convincing when building AI centers in space is access to sunlight (i.e. energy) and the cold of space, which is approximately -200 degrees. With the first, I would not argue; of course, access to sunlight would be available all the time. But the problem is cooling chips and all the infrastructure in space. Why? The answer is: radiators for AI chips, because they cannot be cooled through the circulation of air, as there is empty space, I mean a vacuum. Thus, Musk must send to all its data centers massive radiators that would be cooling AI chips and the AI center.

For example, for 1 MW of AI computing, we need 4 kg of aluminum radiator plate with a thickness of 2–3 mm for each 1 kW of heat. It means that we need around 3–4 tonnes of radiators for 1 MW. It means that for a center with 1 GW, we need 3–4 thousand tonnes of aluminum to be sent to space.

Starship (the goal of SpaceX): Elon Musk has repeatedly said that the target costs are:

- 100–200 USD/kg (very ambitious)
- which means 100–200 thousand USD per tonne

That means 1 billion USD for sending only the radiators to space.

While we need to send the actual AI center, we need to have additional costs of sending H200 servers, solar panels, batteries, construction materials, and robots for easy repairs. It means that, in total, we need to send 50,000 tonnes of equipment to space. The cost of sending it would be astronomical – Falcon Heavy would cost 75 billion USD only to send it to space.

We are not talking about the price of equipment that needs to be paid to NVIDIA and affiliates. It is totally bullshit what Musk is saying in the prospectus. It is achievable, but too expensive.

Correct me if I am wrong.

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Calculation Appendix – 1 MW and 1 GW AI in Space

Assumption 1: in vacuum there is no air circulation, so heat cannot be removed by normal airflow. Heat must be transported from chips to radiator panels and then emitted into space by thermal radiation.

Assumption 2: for a practical low-cost radiator I assume aluminum panels with a black/high-emissivity coating. The aluminum conducts heat inside the panel, while the coating radiates heat into space.

Radiator mass rule used in this article:

4 kg of aluminum radiator structure per 1 kW of heat.

For 1 MW:

1 MW = 1,000 kW

1,000 kW × 4 kg/kW = 4,000 kg

4,000 kg = 4 tonnes of radiator mass

For 1 GW:

1 GW = 1,000 MW = 1,000,000 kW

1,000,000 kW × 4 kg/kW = 4,000,000 kg

4,000,000 kg = 4,000 tonnes of radiator mass

Starship target cost assumption:

100–200 USD/kg

This equals 100,000–200,000 USD per tonne.

Cost of sending only radiators for 1 GW:

4,000 tonnes × 100,000 USD/tonne = 400,000,000 USD

4,000 tonnes × 200,000 USD/tonne = 800,000,000 USD

Rounded conclusion:

Sending only the radiators for a 1 GW orbital AI center could cost around 0.4–0.8 billion USD under very ambitious Starship cost assumptions. With margin, packaging, structure and deployment hardware, this can easily approach around 1 billion USD.

Total AI center mass assumption:

H200 servers + radiators + solar panels + batteries + power electronics + construction + robots + repair equipment = around 50,000 tonnes.

Cost of sending full 1 GW AI center using Starship target cost:

50,000 tonnes × 100,000 USD/tonne = 5,000,000,000 USD

50,000 tonnes × 200,000 USD/tonne = 10,000,000,000 USD

Cost of sending full 1 GW AI center using Falcon Heavy rough cost:

Falcon Heavy rough launch cost used here: 1.5 million USD per tonne.

$50,000 \text{ tonnes} \times 1,500,000 \text{ USD/tonne} = 75,000,000,000 \text{ USD}$

Main calculation conclusion:

Even before buying NVIDIA chips, servers, solar panels, batteries and orbital infrastructure, the logistics cost alone is massive. Under an optimistic Starship scenario it may still be several billions of dollars, while under Falcon Heavy economics it can reach around 75 billion USD just for launch mass.